

Machine Learning Extension of Nunn's Slave Trade and Development Paper

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1 Introduction and Research Question

Double/Debiased Machine Learning (DML) has increasingly been applied to the re-analysis of existing causal studies. It shows flexibility in dealing with the relationships among control variables, treatment variables, and outcome variables. Regardless of whether these relationships between variables are linear, non-linear, or involve complex interaction terms, machine learning methods can be employed to estimate and partial out the components explained by control variables. After residualization process, the treatment and outcome variables used for the final causal estimation are clean and stable, which means they are less influenced by observable confounders. Consequently, compared to traditional linear models, DML reduces bias arising from certain specification and yield more reliable causal estimates.

This report based on Nunn (2008), applying Double/Debiased Machine Learning (DML) methods to the original specification to examine whether casual identification is robust between different learners under the Double/Debiased Machine Learning (DML) framework. Furthermore, this extension investigated whether the core coefficients are significantly different between original results and Double/Debiased Machine Learning (DML) results, and whether there is any improvement after applying Double/Debiased Machine Learning (DML).

Within the DML framework, this extension employs ten different learners, including five linear and five nonlinear learners. The linear learners align with the linear model specification on original study, while nonlinear learners are used to capture nonlinear relationships between control variables and the outcome, thereby checking whether the original functional specifications omit significant nonlinear structures. DML employs 5-fold cross-fitting to reduce the risk of overfitting; and this process is repeated 500 times to avoid sensitivities to certain splits. Finally, the medians of the 500 estimated coefficients and standard errors were selected as the reported results (Chernozhukov et al., 2018).

2 Original Study and Replication

The original paper investigated whether the historical slave trade affects current economic development on African countries. Author used OLS regression, revealing a significantly negative relationship between the intensity of historical slave exports and modern economic development. Across various model specifications with different controls, the estimated coefficients for the slave export variable range from approximately -0.076 to -0.128 . Subsequently, author utilizes instrumental

variables model to address potential endogeneity. The results show that core coefficients range from -0.20 to -0.29 .

Overall, These coefficients can be interpreted as that for every 1% increase in historical slave export intensity, modern per capita income decrease by an average of approximately 0.076% to 0.128% in OLS estimation, while the IV results show a decrease of 0.20% to 0.29%. Overall, the paper concludes that the historical slave trade has a significant and economically meaningful negative impact on the long-term economic development of African countries.

Table 1 and Table 2 report the original results. This extension selects four representative specifications as benchmarks. They are two OLS specifications shown in column (2) and (5) in Table 1, and two IV specifications shown in column (3) and (4) in Table 2. This allows for a comparison of machine learning methods to determine whether using different samples or control variables under the same model specifications yields inconsistent results.

Table 1: Original OLS Estimates from Nunn (2008) Table III

Spec.	Coef.	Added controls/sample	N
(1)	-0.112*** (0.024)	Colonizer FE	52
(2)	-0.076*** (0.029)	+ Geography	52
(3)	-0.108*** (0.038)	Restricted sample	42
(4)	-0.085** (0.035)	+ Island/religion/legal	52
(5)	-0.103*** (0.034)	+ Resource controls	52
(6)	-0.128*** (0.037)	Restricted + resources	42

Notes: The dependent variable is log real GDP per capita in 2000. Coef. is the coefficient on $\ln(\text{exports/area})$; Spec. corresponds to which column in Table III on original paper. standard errors are in parentheses. The dependent variable is log real GDP per capita in 2000. Restricted sample excludes island and North African countries. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

3 Double Machine Learning Framework

This section describes the DML workflow adopted in this report. The DML implementation process can be divided into

Table 2: Original IV Estimates from Nunn (2008) Table IV

Spec.	Coef.	Added controls/sample	N
(1)	-0.208*** (0.053)	No controls	52
(2)	-0.201*** (0.047)	+ Colonizer FE	52
(3)	-0.286* (0.153)	+ Geography	52
(4)	-0.221*** (0.078)	Restricted sample	42

Notes: The dependent variable is log real GDP per capita in 2000. Coef. is the second-stage coefficient on $\ln(\text{exports}/\text{area})$. Standard errors are in parentheses. Instruments are four minimum distances to slave demand markets. Restricted sample excludes island and North African countries. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

four steps. Machine learning methods are employed in the nuisance functions, while the final causal identification is estimated according to the specification in the original study. The work flow includes three conditional expectation models:

$$Y_i = \theta D_i + g(\mathbf{X}_i) + \varepsilon_i, \quad (1)$$

$$D_i = r(\mathbf{X}_i) + v_i, \quad (2)$$

$$Z_i = m(\mathbf{X}_i) + u_i. \quad (3)$$

where, Y_i denotes the outcome variable, D_i denotes the endogenous treatment variable, $\mathbf{X}_i$ is a vector of control variables, and Z_i denotes the instrumental variable.

The workflow as follows:

Step 1: Residualizing the outcome variable Machine-learning methods are first used to estimate the conditional expectation of the outcome:

$$\hat{\mathbb{E}}[Y_i | \mathbf{X}_i]. \quad (4)$$

The residualized outcome is then constructed as:

$$\tilde{Y}_i = Y_i - \hat{\mathbb{E}}[Y_i | \mathbf{X}_i]. \quad (5)$$

In the extension, this part will remove the component of GDP that can be explained by the control variables.

Step 2: Residualizing the treatment variable Next step is used to estimate the conditional expectation of the treatment variable.

$$\hat{\mathbb{E}}[D_i | \mathbf{X}_i]. \quad (6)$$

The residualized treatment variable is:

$$\tilde{D}_i = D_i - \hat{\mathbb{E}}[D_i | \mathbf{X}_i]. \quad (7)$$

For this study, This step removes the component of slave-export intensity that can be explained by the control variables.

Step 3: Residualizing the instrumental variables For each instrumental variable Z_{ij} , machine-learning methods are used to estimate the conditional expectation of the instrumental variables.

$$\hat{\mathbb{E}}[Z_{ij} | \mathbf{X}_i]. \quad (8)$$

The residualized instrumental variables are constructed as:

$$\tilde{Z}_{ij} = Z_{ij} - \hat{\mathbb{E}}[Z_{ij} | \mathbf{X}_i], \quad j = 1, 2, 3, 4. \quad (9)$$

The instrumental variables used in the original study include four minimum distance, distance from the Atlantic, Indian Ocean, Saharan, and Red Sea slave-trade markets. Then the residualized progress should be performed four times.

Step 4: Instrumental-variable estimation using residualized variables For DML-OLS specification, the core coefficient is identified by the following orthogonal moment condition:

$$\mathbb{E}[\tilde{D}_i (\tilde{Y}_i - \theta \tilde{D}_i)] = 0.$$

For DML-IV specification, the first-stage equation is:

$$\tilde{D}_i = \tilde{\mathbf{Z}}_i' \boldsymbol{\pi} + \eta_i. \quad (10)$$

The fitted value from the first stage is:

$$\hat{\tilde{D}}_i = \tilde{\mathbf{Z}}_i' \hat{\boldsymbol{\pi}}. \quad (11)$$

The second-stage equation is:

$$\tilde{Y}_i = \theta \hat{\tilde{D}}_i + \nu_i. \quad (12)$$

Finally, the core coefficient is identified by the following orthogonal moment condition:

$$\mathbb{E}[\tilde{\mathbf{Z}}_i (\tilde{Y}_i - \theta \hat{\tilde{D}}_i)] = 0. \quad (13)$$

4 Machine-Learning Extension and Implementation

In the DML extension analysis, this study retains the original empirical specifications from Nunn (2008), keeping the outcome, treatment, instrument, and control variables unchanged. It should be noted that machine learning methods are not employed to redefine the causal identification for the original study, but to be used to flexibly estimate nuisance functions within the DML framework. Therefore, this section introduces the various learners employed in the extended analysis.

Specifically, this extension selected five linear learners and five non-linear learners. Linear learners consistent with the OLS and IV linear specifications used in the original study and can be compared directly with the original results. In contrast, non-linear learners allow specifications to have non-linear relationships and interactions between the outcome, treatment, instruments, and control variables. By employing learners with different characteristics, this extension examines whether the linear specification in the original study omitted more complex control structures and assesses whether the causal conclusions

results from DML methods are robust and significantly different compared with original conclusions.

The five linear learners are Linear Regression, LASSO, Ridge, Elastic Net, and Polynomial Ridge. Linear Regression employs linear control specification. LASSO used L_1 penalty to shrink the coefficients of some control variables to zero. Therefore, Lasso serves not only as a regularization method but also as a tool for variable selection. Ridge employs an L_2 penalty to shrink the magnitude of the regression coefficients to reduce the impact of noisy or weakly predictive variables of control X . Elastic Net is a combination of Ridge and Lasso with both L_1 and L_2 penalty terms. Consequently, it offers greater flexibility: it can select variables, like Lasso, and shrink coefficients like Ridge. Finally, Polynomial Ridge extends the control variables by adding squared and interaction terms to capture the nonlinear relationship between variables. This approach makes nuisance functions more flexible to exhibit nonlinear structures.

The five non-linear learners are Random Forest, Gradient Boosting, Support Vector Regression (SVR), Extra Tree (Extremely Randomized Trees), and Kernel Ridge.

Random Forest constructs multiple decision trees and averages their predictions to capture non-linear relationships and interactions among the control variables. Extra Tree constructs multiple decision trees as well, and applies a high degree of randomness in that, then averages each prediction. SVR constructs prediction functions to capture non-linear relationships between controls and other variables. Kernel Ridge regression employs kernel functions to original variables and utilizes L_2 regularization for estimation, thereby identifying relatively smooth non-linear relationships. Gradient Boosting builds multiple shallow decision trees, with each new tree correcting errors for the preceding models, thereby progressively improving overall predictive performance.

Table 3 reports the settings for each learner that employed in the extension after considering the sample size and correlation between controls.

In the implementation, this extension employs 5-fold cross-fitting and repeats the random sample 500 times to mitigate the randomness associated with any single split. The final results exact the median of the 500 coefficient estimates and adjust the standard error by incorporating both the uncertainty of individual estimates and the variation across different splits to enhance the stability and robustness of the results.

5 Results and Model Evaluation

5.1 OLS specification

This extension first conducts DML analysis within an OLS framework, comparing column (2) (geographic control model) and column (5) (full controls model) from Table 1 of the original paper. For each model specification, this extension employs ten different learners to residualize the outcome and treatment variables. The coefficient estimates and confidence intervals for each learner are illustrated in Figure 1. Black crosses and horizontal bars represent the estimates and 95% confidence intervals in the original paper, while colored dots represent the median DML estimates from the various learners. It can be seen that all colored dots are below zero and statistically significant.

Table 3: Learners Used in the DML Extension

Learner	Key Settings in Code
Panel A: Linear and Extended Linear Learners	
Linear Regression	<code>StandardScaler() + LinearRegression()</code>
Ridge	<code>StandardScaler() + RidgeCV(alphas=np.logspace(-3, 3, 25))</code>
LASSO	<code>StandardScaler() + LassoCV(cv=3, random_state=610, max_iter=20000)</code>
Elastic Net	<code>StandardScaler() + ElasticNetCV(cv=3, l1_ratio=[0.1, 0.5, 0.9], random_state=610, max_iter=20000)</code>
Polynomial Ridge	<code>StandardScaler() + PolynomialFeatures(degree=2, include_bias=False) + RidgeCV(alphas=np.logspace(-3, 3, 25))</code>
Panel B: Nonlinear Learners	
SVR	<code>StandardScaler() + SVR(C=2.0, epsilon=0.05, gamma="scale")</code>
Kernel Ridge	<code>TransformedTargetRegressor(regressor=StandardScaler() + KernelRidge(alpha=1.0, kernel="rbf"), transformer=StandardScaler())</code>
Random Forest	<code>RandomForestRegressor(n_estimators=35, max_depth=4, max_features="sqrt", min_samples_leaf=3, random_state=610, n_jobs=1)</code>
Extra Trees	<code>ExtraTreesRegressor(n_estimators=35, max_depth=4, max_features="sqrt", min_samples_leaf=3, random_state=610, n_jobs=1)</code>
Gradient Boosting	<code>GradientBoostingRegressor(n_estimators=45, learning_rate=0.04, max_depth=2, min_samples_leaf=3, random_state=610)</code>

This confirms that the negative effect of the slave trade on contemporary income is robust across all specifications, regardless of whether control variables are treated linearly or non-linearly. The confidence intervals for each learner overlap with that of the original study, visually demonstrating there is no significant difference between original results and DML results. For column (2) specification, all coefficients of DML estimations are lower than the original's. For column (5) specification, all coefficients of DML estimations are surrounding with the original's. However, due to the substantial overlap of the confidence intervals, there appear to be no significant differences among the coefficients.

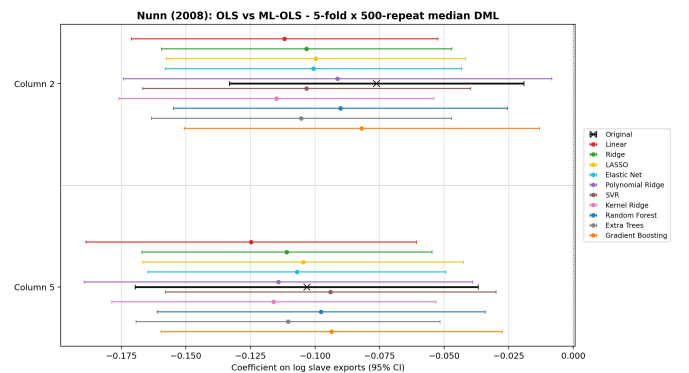


Figure 1: Comparison of OLS and DML-OLS Estimates

Notes: The figure reports the estimated coefficient on the core term, $\ln(\text{exports}/\text{area})$, from each specification. Horizontal lines represent 95% confidence intervals.

Table 4 compares coefficients between original OLS results and DML estimates derived from ten machine learning learn-

Table 4: Original OLS and ML-OLS Estimates

Estimator	Column 2	Column 5
Original	-0.076*** (0.029)	-0.103*** (0.034)
Linear	-0.112*** (0.030)	-0.125*** (0.033)
Ridge	-0.103*** (0.029)	-0.111*** (0.029)
LASSO	-0.100*** (0.030)	-0.105*** (0.032)
Elastic Net	-0.101*** (0.029)	-0.107*** (0.029)
Polynomial Ridge	-0.091** (0.042)	-0.114*** (0.038)
SVR	-0.103*** (0.032)	-0.094*** (0.033)
Kernel Ridge	-0.115*** (0.031)	-0.116*** (0.032)
Random Forest	-0.090*** (0.033)	-0.098*** (0.032)
Extra Trees	-0.105*** (0.030)	-0.110*** (0.030)
Gradient Boosting	-0.082** (0.035)	-0.094*** (0.034)

Notes: The coefficients are derived from the original paper and the DML estimation. Column 2 employs ols specification with geography controls; while Column 5 employs ols specification with full controls. DML use 5-fold cross-fitting repeated 500 times, with one common fold partition shared by all nuisance functions within a repeat. Coefficients are medians across repeats; standard errors were adjusted using the median aggregation of Chernozhukov et al. (2018), $\widehat{se} = \sqrt{\text{med}_r[se_r^2 + (\hat{\theta}_r - \tilde{\theta})^2]}$, combining within-split uncertainty and split variation. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

ers. Under the original specifications, the coefficients for $\ln(\text{exports}/\text{area})$ are -0.076 in column (2) specification and -0.103 in column (5) specification. The estimates from the ten learners, which employs 5-fold cross-fitting and extracts the median of 500 repeated splits, range from -0.082 to -0.125 . All twenty estimates are significantly negative at the 5% level. Table 5 reports the p-value for coefficients equality test, and shows that the DML estimates are statistically insignificant from the original coefficients.

The results in OLS-based estimates reveal two patterns: first, the coefficients from all ten learners in column (2) are more negative than the original estimates, showing a stronger negative effect; second, the original OLS coefficient shifted by 0.027 (approximately 35%) between column (2) and column (5), while the average shift in the machine learning estimates was only 0.009 , indicating a stable negative relationship even though employing different controls.

5.2 IV specification

Figure 2 displays the coefficients for IV specification with column (3) (full sample) and column (4) (restricted sample) from the original paper. The black dots and horizontal bars represent the original IV estimates and 95% confidence intervals, while the colored dots represent the median DML estimates from the various learners. It can be seen that the original confidence interval in column (3) specification across zero, while the intervals for Linear, Ridge, LASSO, Elastic Net, Kernel Ridge, SVR, Gradient Boosting and Extra Trees are below zero, showing statistically significant coefficients. In column (4) specification, where the sample size is reduced to 42 observations, all coefficients of DML estimations are slightly large than the original's, and the confidence intervals for Polynomial Ridge, SVR, Random Forest, Extra Trees, and Gradient Boosting cross the zero line with negative coefficients, while intervals for other

Table 5: Tests of $H_0 : \theta_{ML} = \theta_{Orig}$, OLS Specifications

<i>Panel A: OLS Specification, Column 2</i>			
Learner	Difference	SE	p-value
<i>Benchmark: $\hat{\theta}_{Orig} = -0.076$ (0.029)</i>			
Linear	-0.036	0.042	0.391
Ridge	-0.027	0.041	0.510
LASSO	-0.024	0.041	0.558
Elastic Net	-0.024	0.041	0.558
Polynomial Ridge	-0.015	0.051	0.769
SVR	-0.027	0.044	0.540
Kernel Ridge	-0.039	0.043	0.364
Random Forest	-0.014	0.044	0.750
Extra Trees	-0.029	0.042	0.490
Gradient Boosting	-0.006	0.046	0.896
<i>Panel B: OLS Specification, Column 5</i>			
Learner	Difference	SE	p-value
<i>Benchmark: $\hat{\theta}_{Orig} = -0.103$ (0.034)</i>			
Linear	-0.022	0.047	0.640
Ridge	-0.008	0.044	0.856
LASSO	-0.001	0.046	0.983
Elastic Net	-0.004	0.045	0.929
Polynomial Ridge	-0.011	0.051	0.829
SVR	0.009	0.047	0.848
Kernel Ridge	-0.013	0.047	0.782
Random Forest	0.006	0.047	0.898
Extra Trees	-0.007	0.045	0.876
Gradient Boosting	0.010	0.048	0.835

Notes: The table reports tests of $H_0 : \theta_{ML} = \theta_{Orig}$. The coefficients are derived from the original paper and the DML estimation. Difference denotes $\hat{\theta}_{ML} - \hat{\theta}_{Orig}$, and SE is the pooled standard error $\sqrt{SE_{ML}^2 + SE_{Orig}^2}$. The two-sided p-value is calculated from the standard normal statistic

$$z = \frac{\hat{\theta}_{ML} - \hat{\theta}_{Orig}}{\sqrt{SE_{ML}^2 + SE_{Orig}^2}}.$$

$\hat{\theta}_{Orig}$ is the replicated estimate from the corresponding published specification. SE_{ML} is the median-aggregated DML standard error across 500 repetitions. Because both estimates use the same country sample, their covariance is likely positive. Setting this covariance to zero increases the estimated standard error of the difference and therefore makes the test conservative.

learners are significant.

Overall, the magnitude for all twenty learners are close to the original estimate and cluster around the original point estimate. Therefore, the results are indicate a robust relationship between modern economic developments and slave export intensity $\ln(exports/area)$, indicating that the original findings were not affected by nonlinear structures among the variables. However, due to the substantial overlap of the confidence intervals, there appear to be no significant differences among the coefficients.

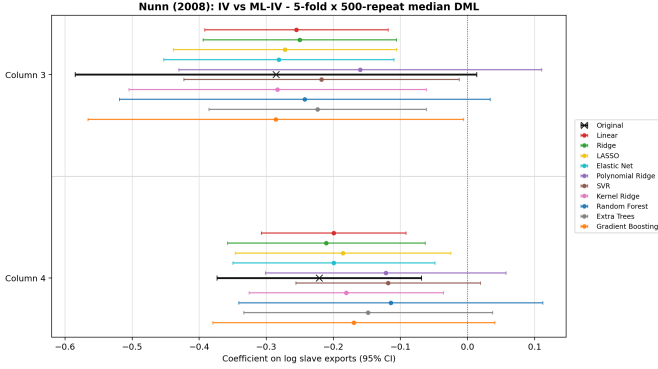


Figure 2: Comparison of IV and DML-IV Estimates

Notes: The figure reports the estimated coefficient on the core term, $\ln(exports/area)$, from each specification. Horizontal lines represent 95% confidence intervals.

Table 6 compares the original 2SLS estimates with DML-IV estimates using ten different machine learning learners; and the instrumental variables are four minimum distance to the slave demand markets. Under the original specification, the core coefficient in column (3) is 0.286 with marginally significant at the 10% level, while in column (4) it is 0.221 with significant at the 5% level. Across the ten machine learning methods, the estimates range from -0.160 to -0.286 for column (3) (full sample) and from -0.114 to -0.211 for column (4) (restricted sample). All estimates are negative, keeping the same sign with the original estimates. Table 7 shows none of the twenty estimates differ significantly from the original coefficients.

Notably, there are all learners in column (3) with lower standard errors than the original; specifically, Linear regression is 0.070, Ridge is 0.074, LASSO is 0.085, and Elastic Net is 0.088, achieving reductions of 42% to 56% compared to the originals. Those improvement elevated the estimates from marginally significant to strong significance at the 1% level. This improvement in precision can be attributed to machine learning methods that residualized outcome variables against control variables and reduce the residual component in the IV variance formula, rather than strengthening the instrument itself.

6 Economic Interpretation and Limitations

In terms of economic interpretation, the results of DML can be summarized as follows: these estimates confirm the nearly magnitude of the negative effect of slave export that estimated

Table 6: Original IV and ML-IV Estimates

Estimator	Column 3	Column 4
Original	-0.286* (0.153)	-0.221*** (0.078)
Linear	-0.255*** (0.070)	-0.200*** (0.055)
Ridge	-0.250*** (0.074)	-0.211*** (0.075)
LASSO	-0.272*** (0.085)	-0.186** (0.082)
Elastic Net	-0.281*** (0.088)	-0.199*** (0.077)
Polynomial Ridge	-0.160 (0.138)	-0.122 (0.092)
SVR	-0.218** (0.105)	-0.119* (0.070)
Kernel Ridge	-0.283** (0.113)	-0.181** (0.074)
Random Forest	-0.243* (0.141)	-0.114 (0.116)
Extra Trees	-0.223*** (0.083)	-0.148 (0.095)
Gradient Boosting	-0.286** (0.143)	-0.170 (0.107)

Notes: Entries are coefficients on log slave exports per unit land area. Original rows reproduce Nunn (2008) with the conventional standard errors printed in the paper. ML rows use 5-fold cross-fitting repeated 500 times, with one common fold partition shared by all nuisance functions within a repeat. Coefficients are medians across repeats; standard errors use the median aggregation of Chernozhukov et al. (2018), $\hat{se} = \sqrt{\text{med}_r[se_r^2 + (\hat{\theta}_r - \tilde{\theta})^2]}$, combining within-split uncertainty and split variation. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

in the original study, furthermore, they narrow the range of uncertainty included in that effect.

For example, in Column 3, the estimate points cluster between 0.25 and 0.29, which is relatively close to the original estimate of 0.286. Those coefficients can be interpreted as: a 10% increase in the intensity of slave exports is associated with an average decline of about 2.5% to 2.9% in per capita income in the year 2000. Considering the vast disparities in exposure to the slave trade among the countries in the sample, this elasticity can translate into substantial income gaps.

This extension has several limitations. First, DML methods improve the estimation process rather than the causal identification. It allows control variables to enter the model in a more flexible manner, while it still cannot exclude the influence of unobserved factors. Regarding implementation, DML methods introduce additional choices when conducting, such as different learners, number of folds, number of repetitions, and aggregation method. However, the different settings may yield varying results. Although this extension conduct 500 repetitions, uses median aggregation, and reports results for ten different learners, the issue cannot be entirely eliminated. Furthermore, in IV models, DML does not resolve the weak instrument problem. In this extension, F-statistic have not improved in the first-stage relationship.

7 Conclusion

Building on Nunn (2008), this extension employed Double Machine Learning (DML) framework to re-estimate the causal effect of the slave trade on per capita GDP in the year 2000. The analysis employs ten learners, including five linear and five nonlinear learners, with 5-fold cross-fitting, 500 repeated splits, and median aggregation. This setting flexibly capture the relationships across variables and reduce the risk of specification bias and overfitting issues.

Table 7: Tests of $H_0 : \theta_{ML} = \theta_{Orig}$, IV Specifications

<i>Panel A: IV Specification, Column 3</i>			
Learner	Difference	SE	p-value
<i>Benchmark: $\hat{\theta}_{Orig} = -0.286$ (0.153)</i>			
Linear	0.031	0.168	0.854
Ridge	0.035	0.170	0.837
LASSO	0.013	0.175	0.941
Elastic Net	0.004	0.176	0.982
Polynomial Ridge	0.126	0.206	0.541
SVR	0.068	0.185	0.713
Kernel Ridge	0.002	0.190	0.992
Random Forest	0.043	0.208	0.836
Extra Trees	0.062	0.174	0.722
Gradient Boosting	-0.000	0.209	1.000
<i>Panel B: IV Specification, Column 4</i>			
Learner	Difference	SE	p-value
<i>Benchmark: $\hat{\theta}_{Orig} = -0.221$ (0.078)</i>			
Linear	0.022	0.095	0.817
Ridge	0.011	0.108	0.919
LASSO	0.035	0.113	0.757
Elastic Net	0.022	0.109	0.840
Polynomial Ridge	0.099	0.120	0.409
SVR	0.103	0.105	0.327
Kernel Ridge	0.040	0.107	0.709
Random Forest	0.107	0.139	0.441
Extra Trees	0.073	0.123	0.553
Gradient Boosting	0.052	0.132	0.694

Notes: The table reports tests of $H_0 : \theta_{ML} = \theta_{Orig}$. The coefficients are derived from the original paper and the DML estimation. Difference denotes $\hat{\theta}_{ML} - \hat{\theta}_{Orig}$, and SE is the pooled standard error $\sqrt{SE_{ML}^2 + SE_{Orig}^2}$. The two-sided p-value is calculated from the standard normal statistic

$$z = \frac{\hat{\theta}_{ML} - \hat{\theta}_{Orig}}{\sqrt{SE_{ML}^2 + SE_{Orig}^2}}.$$

$\hat{\theta}_{Orig}$ is the replicated estimate from the corresponding published specification. SE_{ML} is the median-aggregated DML standard error across 500 repetitions. Because both estimates use the same country sample, their covariance is likely positive. Setting this covariance to zero increases the estimated standard error of the difference and therefore makes the test conservative.

The findings conclude that the estimates derived from the DML framework consistent with the original results in sign, and there is no statistically significant difference between the DML coefficients and original results. All of OLS-based estimates under DML methods are significant at the 5% level; furthermore, there is no significant difference between linear and non-linear learners, indicating the original results are robust regardless of the choice of functional form. In the IV-based estimates, machine learning substantially improves the quality of inference: the standard errors in some learners on the column (3) (IV specification) are reduced by 42%–56% compared to the original estimates, improving the coefficients from marginally significant to strongly significant in some DML estimations. This improvement stems from the residualization of variables rather than from an enhancement of the instrument itself.

Overall, DML methods do not alter the original core conclusions but reinforces its robustness by employing various learners that can effectively handle high-dimensional or correlation variables involving controls. There is no significant difference between original coefficients and DML estimate coefficients. The conclusion of the original paper does not relay entirely on a simple linear control specification. Even when allowing for more complex, non-linear relationships between the control variables and income, slave exports, and instrumental variables, the negative impact of the slave trade remains robust. Furthermore, in some models, the DML approach narrows the confidence intervals and improved the accuracy of estimation.

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